

EXTENSION 5E Adding and Subtracting Algebraic Fractions

Learning intentions

- To understand what an algebraic fraction is
- To be able to find the lowest common denominator of two algebraic fractions
- To be able to find equivalent algebraic fractions with different denominators
- To be able to add and subtract algebraic fractions and simplify the result

Key Ideas

- An **algebraic fraction** is a fraction with an algebraic expression as the numerator or the denominator.
- The **lowest common denominator** (or LCD) of two algebraic fractions is the smallest multiple of the denominators.
- Adding and subtracting algebraic fractions requires that they both have the same denominator.
For example: $\frac{2x}{5} + \frac{4y}{5} = \frac{2x + 4y}{5}$

Example 8 | Working with denominators

- Find the lowest common denominator of $\frac{3x}{10}$ and $\frac{2y}{15}$.
- Convert $\frac{2x}{7}$ to an equivalent algebraic fraction with the denominator 21.

$$\begin{aligned} \text{a) } & 10, 20, 30, \dots \\ & 15, 30, \dots \\ & \text{LCD} = 30 \end{aligned}$$

$$\text{b) } \frac{2x}{7} \times 3 \quad ; \quad \frac{6x}{21}$$

Example 9 | Adding and subtracting algebraic fractions

Simplify the following expressions.

a $\frac{3x}{11} + \frac{5x}{11}$

$$= \frac{8x}{11}$$

b $\frac{4a}{3} + \frac{2a}{5}$

$$= \frac{4a \times 5 + 2a \times 3}{3 \times 5}$$

$$= \frac{20a + 6a}{15}$$

$$= \frac{26a}{15}$$

c $\frac{6k}{5} - \frac{3k}{10}$

$$= \frac{6k \times 10 - 3k \times 5}{5 \times 10}$$

$$= \frac{60k - 15k}{50}$$

$$= \frac{45k}{50} \div 5$$

$$= \frac{9k}{10}$$

d $\frac{a}{6} - \frac{b}{9}$

$$= \frac{a \times 3 - b \times 2}{54}$$

$$= \frac{3a - 2b}{54} \div 3$$

$$= \frac{3a - 2b}{18}$$



x



x